REPEATED DEGREES IN FINITE SIMPLE GRAPHS

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Abstract. It is a well-known party puzzle that among any two or more people, two have the same number of acquaintances present. We give a careful proof of this fact for finite simple graphs, based on the observation that a graph cannot contain both an isolated and a dominating vertex. We also collect two elementary sufficient conditions for connectivity and comment on the characterisation of degree sequences due to Erdős and Gallai.

1. Introduction

The pigeonhole argument showing that no finite simple graph on at least two vertices has all degrees distinct appears in many textbooks, e.g. [1]. Our goal is to present it (Theorem 2.6) in a form where every step is isolated as a separate statement, together with some related facts on degrees and connectivity that are frequently used alongside it.

2. Degrees

Definition 2.1. A *graph* $G = (V, E)$ consists of a finite set V of *vertices* and a set E of 2-element subsets of V, the *edges*. We write $n = |V|$. For $v \in V$, the *neighbourhood* of v is $N(v) = \{u \in V : \{u, v\} \in E\}$ and the *degree* of v is $\deg v = |N(v)|$. A vertex is *isolated* if its degree is 0 and *dominating* if its degree is $n - 1$. The graph is *connected* if any two vertices are joined by a path.

Lemma 2.2 (Handshake lemma)**.** *For every graph G, $\sum_{v \in V} \deg v = 2|E|$.*

Proof. Count the pairs (v, e) with $v \in e \in E$: each edge contains exactly two vertices, and each vertex v lies in exactly $\deg v$ edges. □

Corollary 2.3. *In every graph the number of vertices of odd degree is even.*

Proof. Reduce Lemma 2.2 modulo 2: the sum of the odd degrees is congruent to the number of vertices of odd degree, and it is even. □

Lemma 2.4. *For every vertex v of a graph on n vertices, $0 \leq \deg v \leq n-1$.*

Proof. $N(v) \subseteq V \setminus \{v\}$, since edges have two distinct elements. □

Lemma 2.5. *No graph has both an isolated vertex and a dominating vertex.*

Proof. Suppose u is dominating and w is isolated. Then $u \neq w$, because their degrees $n - 1$ and 0 differ. As u is adjacent to every other vertex, $w \in N(u)$, so $u \in N(w)$ and $\deg w \geq 1$, a contradiction. □

Theorem 2.6 (Main theorem)**.** *Let G be a graph with $n \geq 2$ vertices. Then there are two distinct vertices $u \neq w$ with $\deg u = \deg w$.*

Proof. By Lemma 2.4 every degree lies in $\{0, 1, \ldots, n-1\}$, a set of n values. By Lemma 2.5 the values 0 and $n-1$ do not both occur, so the degrees of G take at most $n-2$ distinct values. Since there are n vertices, the pigeonhole principle gives two distinct vertices with the same degree. □

Proposition 2.7. *Let G be a graph on $n \geq 2$ vertices whose degrees are pairwise distinct. Then G has exactly one isolated vertex and exactly one dominating vertex.*

Proof. The n distinct degrees lie in the n-element set $\{0, \ldots, n-1\}$ by Lemma 2.4, so each value is taken exactly once; in particular 0 and $n-1$ are each taken by exactly one vertex. □

3. Connectivity

Proposition 3.1. *Let G be a graph on $n \geq 1$ vertices in which every vertex has degree at least $\frac{n-2}{2}$. Then G is connected.*

Proof. Let $u \neq w$ be non-adjacent vertices. Then $N(u)$ and $N(w)$ are subsets of $V \setminus \{u, w\}$, which has $n-2$ elements, while $|N(u)| + |N(w)| \geq \frac{n-1}{2} + \frac{n-1}{2} = n-1 > n-2$. Hence u and w have a common neighbour, so any two vertices are joined by a path of length at most 2. □

Proposition 3.2. *Let G be a graph on $n \geq 1$ vertices with $|E| \geq \binom{n-1}{2}$. Then G is connected.*

Proof. If G is not connected, split $V = A \sqcup B$ with A, B nonempty and no edge between A and B. With $a = |A|$ we get $|E| \leq \binom{a}{2} + \binom{n-a}{2} \leq \binom{n-1}{2}$, since the middle expression is a convex function of $a \in [1, n-1]$, maximised at the endpoints. □

4. Degree sequences

The *degree sequence* of G is the list of its degrees in non-increasing order. We recall the classical characterisation of degree sequences.

Theorem 4.1 (Erdős–Gallai [2])**.** *A sequence $d_1 \geq d_2 \geq \cdots \geq d_n \geq 0$ of integers is the degree sequence of a graph if and only if $\sum_{i=1}^{n} d_i$ is even and, for every $1 \leq k \leq n$,*

$$\sum_{i=1}^{k} d_i \;\leq\; k(k-1) + \sum_{i=k+1}^{n} d_i.$$

Remark. By Theorem 2.6, the sequence $(3, 2, 1, 0)$ is not a degree sequence; consistently, it violates the inequality of Theorem 4.1 at $k = 2$, since $5 > 2 + 1$.

References

[1] B. Bollobás, *Modern Graph Theory*, Graduate Texts in Mathematics 184, Springer, 1998.

[2] P. Erdős and T. Gallai, *Gráfok előírt fokú pontokkal*, Mat. Lapok **11** (1960), 264–274.

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